

TALHA SAJID

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[LinkedIn](#) | [Portfolio](#) |

PROFESSIONAL PROFILE

CFD Engineer with a Master's degree in Applied Computational Mechanics and an MPhil in Mathematics. Strong academic foundation in computational fluid dynamics, mathematical modeling, numerical methods, simulation, and engineering data analysis. Experienced in applying CFD and scientific computing techniques to analyze fluid flow, heat transfer, turbulence related behavior, and thermal fluid engineering problems. Focused on using simulation and mathematical modeling to support practical engineering analysis and problem solving. Seeking CFD Engineer, R&D Engineer, Simulation Engineer roles in engineering, energy, industrial, and research driven environments.

EDUCATION

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| Skolkovo Institute of Science and Technology, Skoltech
<i>Research Master of Applied Computational Mechanics; CGPA: 3.8/4.0</i> <ul style="list-style-type: none">Research focus: computational fluid dynamics, heat and mass transfer, multiphase flow, turbulence modeling, numerical simulation, mathematical modeling, and engineering applications in industrial systems, and renewable energy. | Moscow, Russia
Sep. 2024 – Jun. 2026 |
| Government College University Lahore
<i>MPhil in Mathematics</i> <ul style="list-style-type: none">Academic focus: Nonlinear Systems, Numerical Methods for PDEs, Special Functions, Chaos Theory Mathematical Modeling. | Lahore, Pakistan
Nov. 2021 – Dec. 2023 |
| University of the Punjab
<i>BS in Mathematics</i> <ul style="list-style-type: none">Academic focus: Calculus I and II, Differential Equations (ODEs and PDEs), Complex Analysis, Numerical Methods, Computational Mathematics, Statistics. | Lahore, Pakistan
Nov. 2017 – Jun. 2021 |

WORK EXPERIENCE

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| Computational Engineer
<i>Tekvel</i> <ul style="list-style-type: none">Supported engineering research through computational modeling, process-control analysis, and simulation-based evaluation of technical systems.Applied numerical methods to study fluid-flow, heat-transfer, and mass-transfer behavior for process performance and engineering optimization. | Jan. 2026 – May . 2026
Moscow, Russia |
| CFD Engineer
<i>Eco Energy LLC</i> <ul style="list-style-type: none">Performed CFD simulations and numerical modeling to support renewable energy research focused on biogas systems.Collaborated with the research team to analyze flow behavior, thermal performance, and engineering parameters for sustainable energy applications. | Apr. 2025 – Oct. 2025
Moscow, Russia – Hybrid |
| Trainee Engineer
<i>Automate Engineering And Services</i> <ul style="list-style-type: none">Investigated computational analysis and CFD simulations for engineering systems, focusing on fluid flow, heat transfer, and performance optimization.Supported project teams through model setup, mesh preparation, solver execution, post-processing, and clear reporting of simulation results. | Mar. 2022 – Sep. 2024
Lahore, Pakistan |

TECHNICAL SKILLS

Programming: Python, MATLAB, C++, C, Java,
Framework: PyTorch, TensorFlow, OpenCV, NumPy, SciPy, Pandas, Matplotlib
Simulation Tools: ANSYS Fluent, OpenFOAM, SolidWorks, Abaqus, AutoCAD, MUFiT
Tools and Environments: Git/GitHub, Jupyter, Linux, L^AT_EX, VS Code

RESEARCH EXPERIENCE

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| Master's Thesis: Numerical Investigation of Supersonic Reactive and Non-reactive Flow Around a Heated Cylinder
<i>Skoltech, Moscow, Russia</i> <ul style="list-style-type: none">Investigated high-speed aerodynamic and thermal flow phenomena through CFD modeling of reactive and non-reactive supersonic flows around a heated cylinder.Used ANSYS Fluent and OpenFOAM to compare shock-wave behavior, wake development, pressure, and temperature fields.Studied the effects of Mach number, cylinder diameter, wall heating, and hydrogen-air combustion on flow performance. | Sep. 2025 – Jun. 2026 |
| Improved Approach for Implementing Dynamic Mode Decomposition with Control
<i>Skoltech, Moscow, Russia</i> <ul style="list-style-type: none">Implemented Dynamic Mode Decomposition with Control (DMDc) in Python to model and reconstruct nonlinear systems, including Lorenz and Lotka–Volterra dynamics. | Mar. 2025 – Jul. 2025 |

- Applied SVD, low-rank approximation, eigenvalue analysis, and trajectory reconstruction to evaluate model accuracy, stability, and computational efficiency.

Numerical Modeling of Shock Wave Interactions in a Heated 1D Domain

Skoltech, Moscow, Russia

Mar. 2025 – Jun. 2025

- Modeled 1D shock-wave interaction with a heated region using finite-volume solutions of the compressible Euler equations.
- Benchmarked Lax–Friedrichs and HLLC solvers against an exact Riemann solution to assess accuracy, diffusion, and discontinuity resolution.

Similarity Based Modeling of Thin-Film Spreading and Laminar Boundary Layers

Skoltech, Moscow, Russia

Jan. 2025 – Mar. 2025

- Analyzed thin-film flow, boundary-layer behavior, and flat-plate drag using Reynolds lubrication theory, similarity solutions, and Blasius flow analysis.
- Applied analytical and numerical methods to estimate dimensionless parameters, solve boundary-layer problems, and evaluate laminar flow velocity profiles.

Iterative Numerical Modeling of 2D Elliptic PDEs

Skoltech, Moscow, Russia

Jan. 2024 – Mar. 2025

- Developed a finite-difference solution for the 2D Poisson equation with Dirichlet boundary conditions and verified second-order accuracy.
- Compared SOR and Conjugate Gradient methods by evaluating convergence rate, iteration count, grid sensitivity, and maximum error.

Numerical Methods for ODE-Based Engineering and Dynamical Systems

Skoltech, Moscow, Russia

Oct. 2024 – Feb. 2025

- Applied numerical methods, including RK4, Adams-Bashforth, Leapfrog, Euler, and RK2 schemes, to solve nonlinear ODEs and engineering problems.
- Evaluated numerical accuracy, stability, and dynamical behavior through applications in ice growth, predator-prey systems, and nonlinear oscillators.

MPhil Thesis: Mathematical Model of the Coolant Flow in Cylindrical Tubes in Automobile Radiators

Government College University Lahore (GCUL), Lahore, Pakistan

Sep. 2022 – Jul. 2023

- Developed analytical solutions for non-Newtonian heat and mass transfer models to compare automotive coolant performance and identify ethylene glycol as the most effective option.

CONFERENCES AND SEMINARS

Research Seminars in Applied Computational Mechanics

Skolkovo Institute of Science and Technology, Moscow, Russia

Nov. 2024 – May 2025

- Participated in advanced seminars covering recent developments in computational mechanics.

Research Presentation: Mathematical Modeling of Coolant Flow in Automobile Radiator Tubes

4th International Conference on Computing, Mathematics, and Engineering Technologies (iCoMET)

Feb. 2023

Sukkur IBA University, Pakistan

- Presented research on heat and mass transfer in non-Newtonian coolant flow, using Laplace and finite Hankel transforms to solve governing equations and evaluate automotive radiator coolant performance.

Workshop Participation: Advancements in Mathematics and Its Applications

First Workshop on Advancements in Mathematics & Its Application

May 2023

Riphah International University, Lahore, Pakistan

- Attended technical sessions on recent developments, methods, and applications in applied mathematics.

Conference Participation: Mathematical Modeling and Scientific Computing

International Conference on Mathematical Modeling and Scientific Computing for Industrial Research

Nov. 2022

Abdus Salam School of Mathematical Sciences, Lahore, Pakistan

- Participated in sessions focused on mathematical modeling, scientific computing, and industrial research applications.

AWARDS AND SCHOLARSHIPS

Skoltech Merit Scholarship

Skolkovo Institute of Science and Technology (Skoltech), Moscow, Russia

Sep. 2024 – Jun. 2026

- Awarded a merit-based full scholarship in recognition of strong academic performance and potential in computational mechanics.

Special Person Merit Scholarship

University of the Punjab, Lahore, Pakistan

Nov. 2017 – Jun. 2021

- Received a merit-based scholarship in recognition of academic excellence during undergraduate studies.

LANGUAGES

English: C1

Urdu: Native

ACADEMIC REFERENCES

Prof. Aslan R. Kasimov

Research Advisor, Master in Applied Computational Mechanics

Skolkovo Institute of Science and Technology, Moscow, Russia

Email: A.Kasimov@skoltech.ru

Dr. Azhar Ali Zafar

Supervisor, Department of Mathematics

Government College University Lahore (GCUL), Lahore, Pakistan

Email: azharalizafar@gcu.edu.pk